



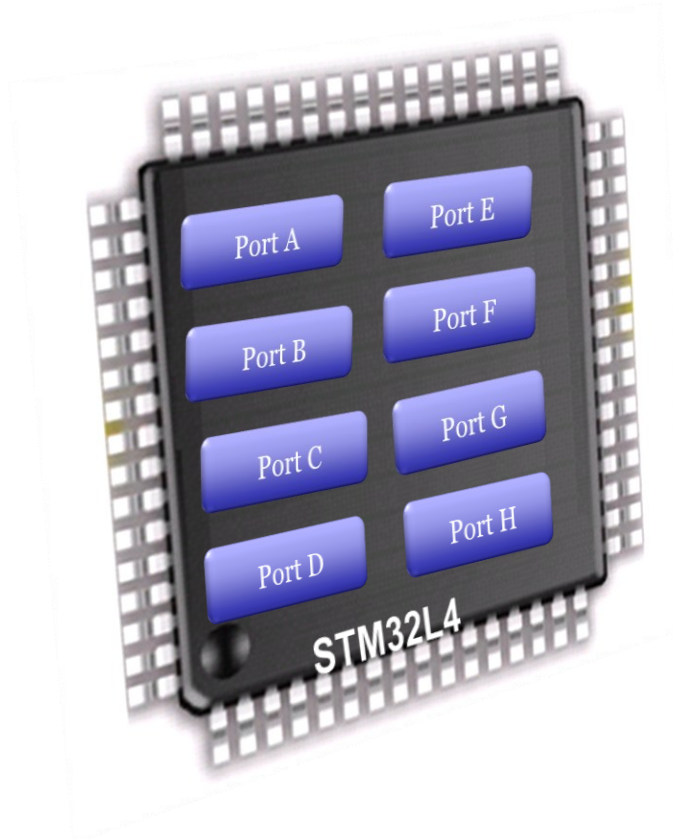
Sichuan University -
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ECE0202 Embedded Processors & Interfacing + Lab

Sichuan University-Pittsburgh Institute



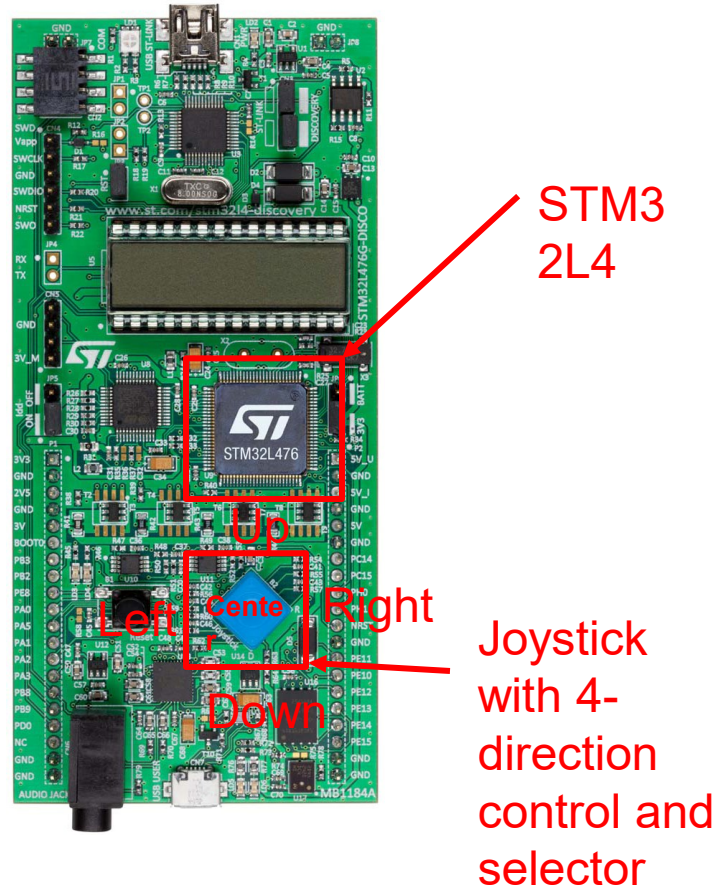
General Purpose Input/Output (GPIO)



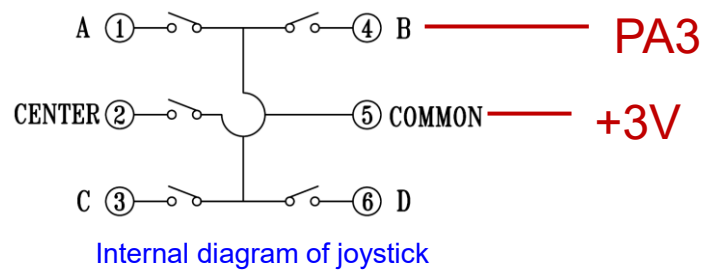
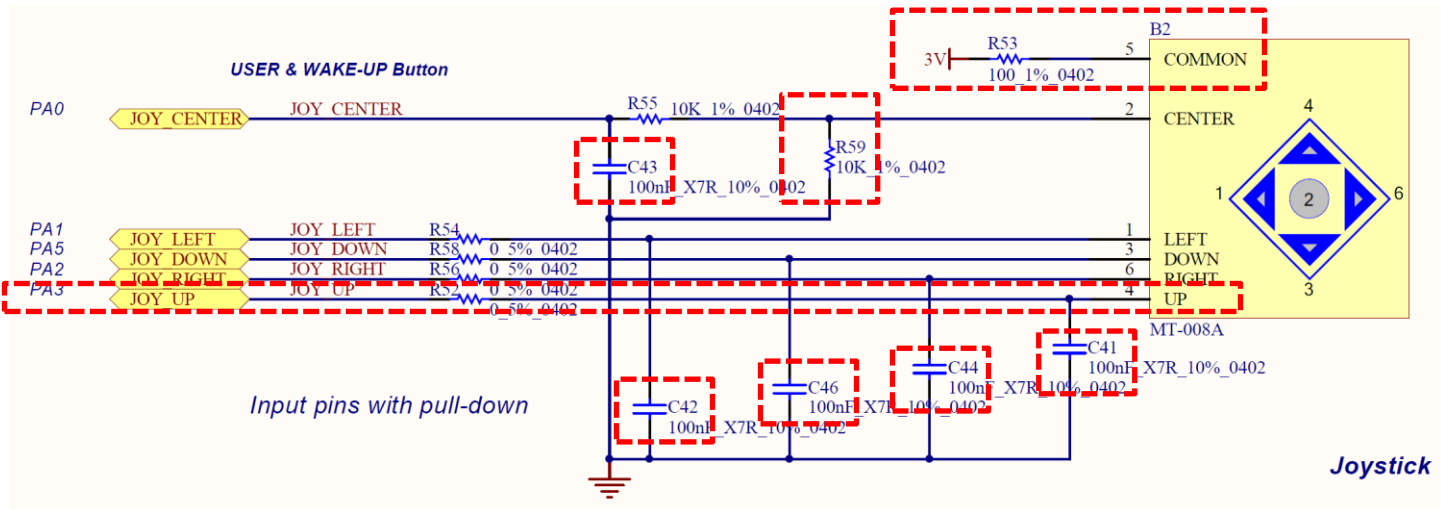
- 8 GPIO Ports:
A, B, C, D, E, F, G, H
- Up to 16 pins in each port

Joystick

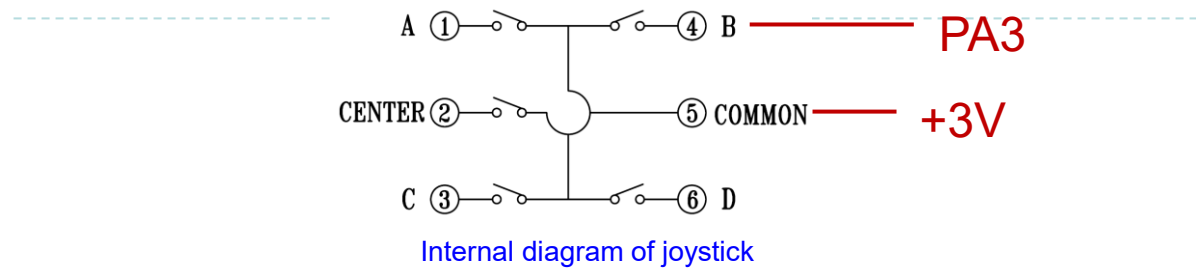
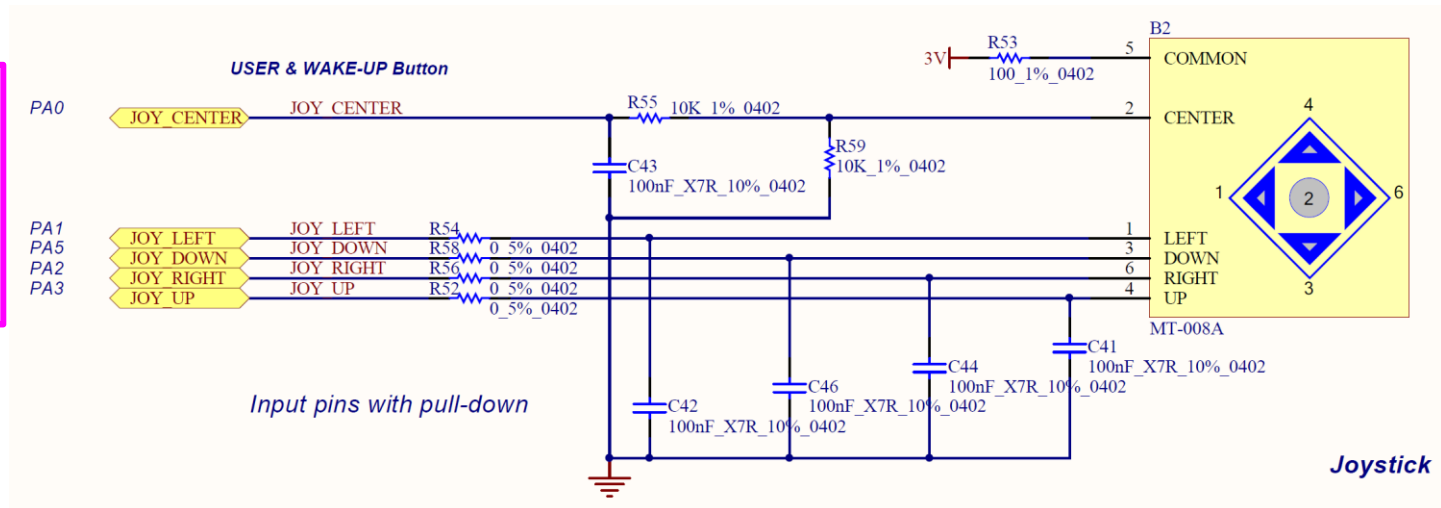
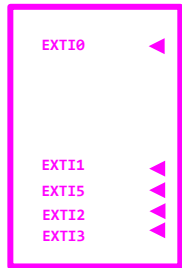
STM32L4 Discovery Kit



Joystick



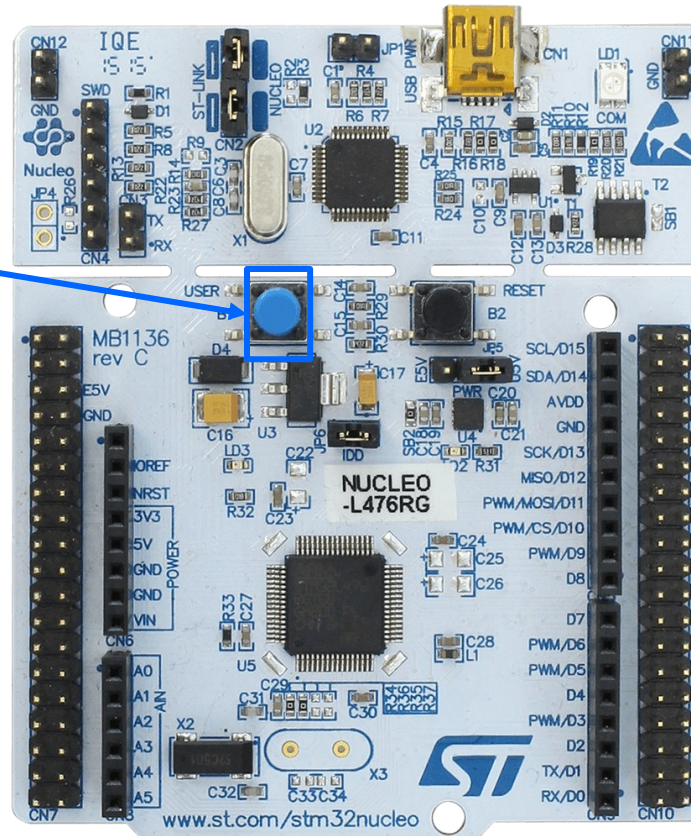
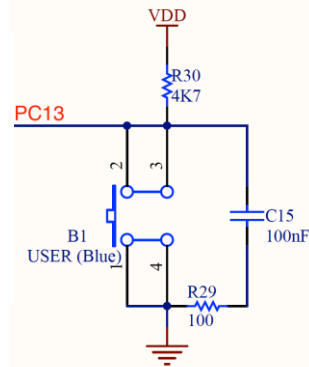
Joystick



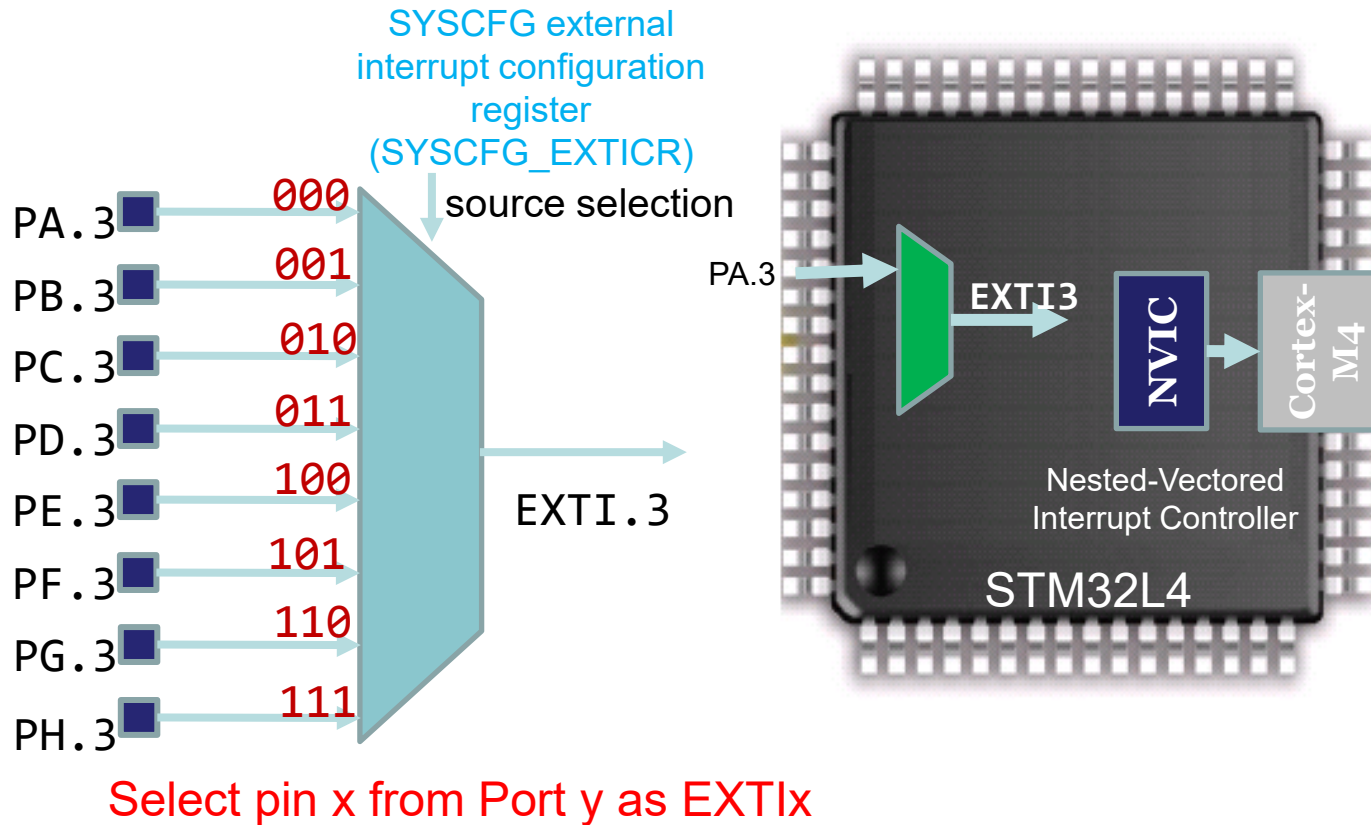
Push Button



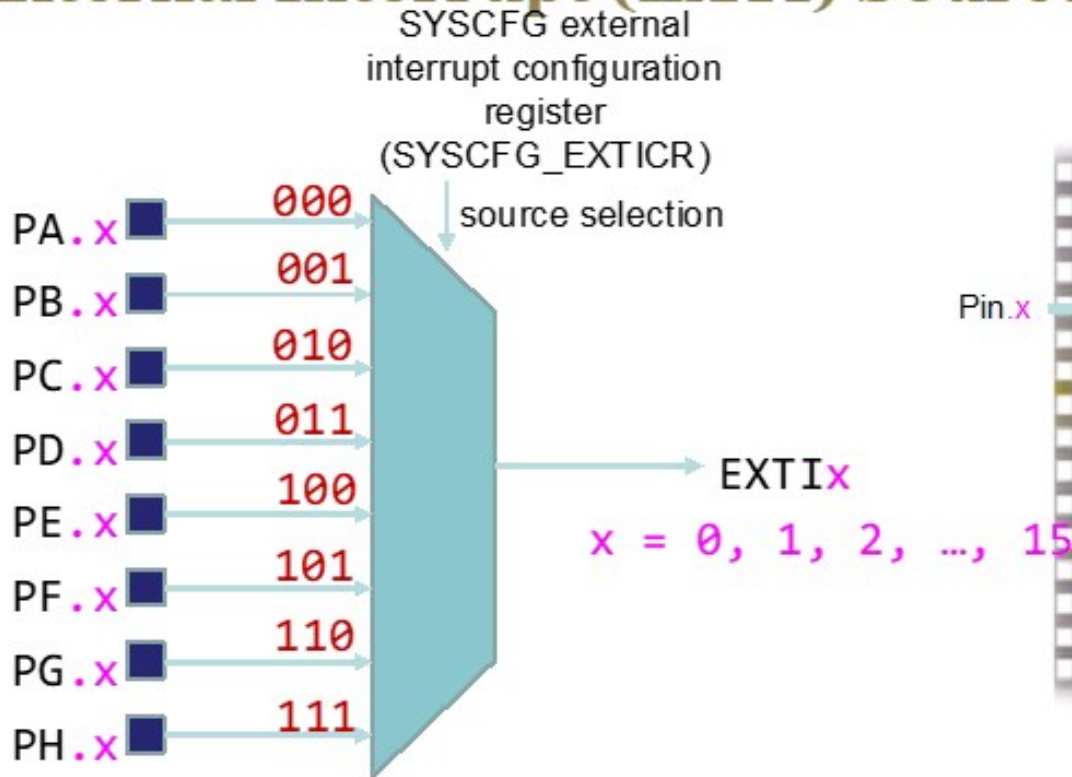
Push Button



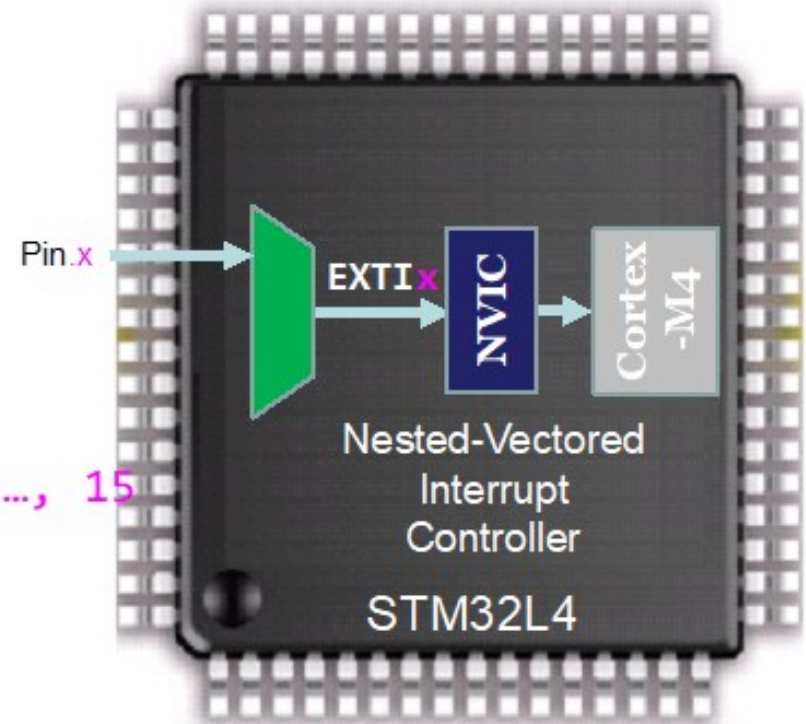
External Interrupt (EXTI) Sources

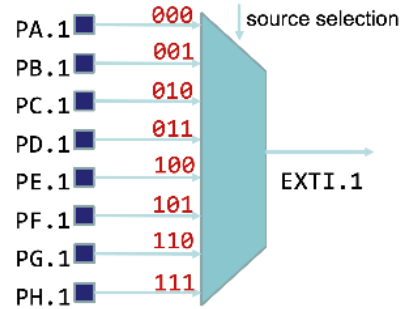
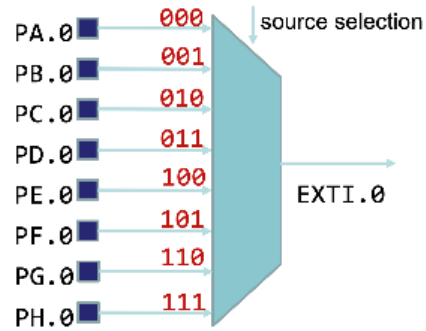


External Interrupt (EXTI) Sources

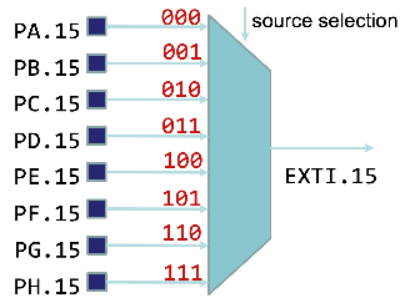


One mux for each of EXTI0, EXTI1, ..., EXTI15





...



SYSCFG_EXTICR1 - 4 (External Interrupt Configuration Register)

SYSCFG_EXTICR1

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res	EXTI3[2:0]			Res	EXTI2[2:0]			Res	EXTI1[2:0]			Res	EXTI0[2:0]		
	r/w	r/w	r/w		r/w	r/w	r/w		r/w	r/w	r/w		r/w	r/w	r/w

Bits 31:15 Reserved, must be kept at reset value.

Bits 14:12 **EXTI3[2:0]**: EXTI 3 configuration bits

These bits are written by software to select the source input for the EXTI3 external interrupt.

000: PA[3] pin

001: PB[3] pin

010: PC[3] pin

011: PD[3] pin

100: PE[3] pin

101: PF[3] pin

110: PG[3] pin

111: Reserved

SYSCFG_EXTICR1 - 4 (External Interrupt Configuration Register)

SYSCFG_EXTICR2

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res	EXTI7[2:0]			Res	EXTI6[2:0]			Res	EXTI5[2:0]			Res	EXTI4[2:0]		
	r/w	r/w	r/w		r/w	r/w	r/w		r/w	r/w	r/w		r/w	r/w	r/w

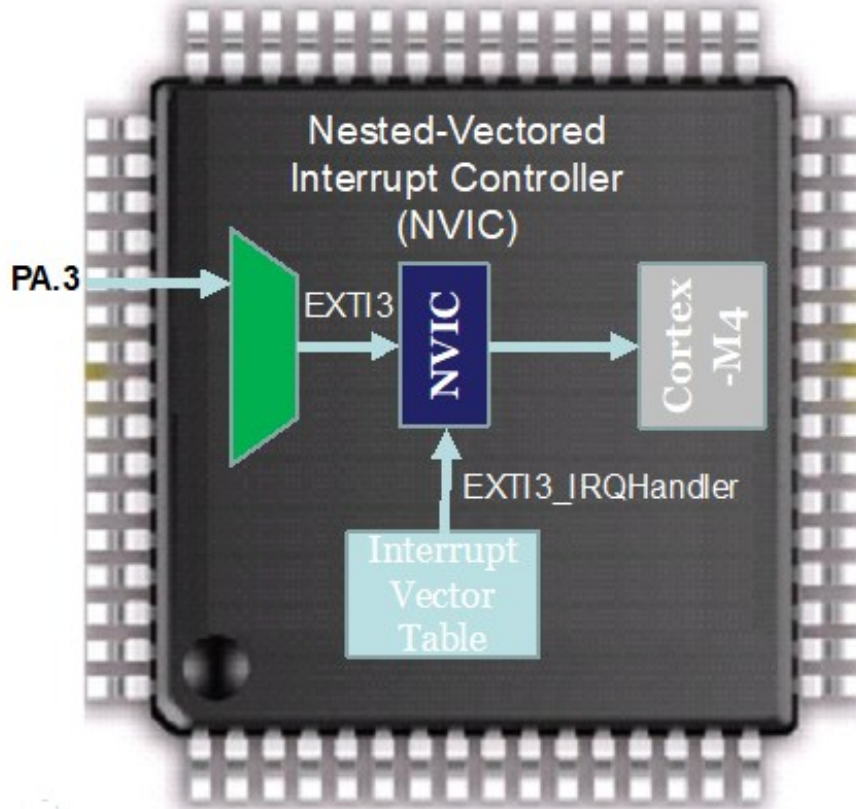
SYSCFG_EXTICR3

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res	EXTI11[2:0]			Res	EXTI10[2:0]			Res	EXTI9[2:0]			Res	EXTI8[2:0]		
	r/w	r/w	r/w		r/w	r/w	r/w		r/w	r/w	r/w		r/w	r/w	r/w

SYSCFG_EXTICR4

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res	EXTI15[2:0]			Res	EXTI14[2:0]			Res	EXTI13[2:0]			Res	EXTI12[2:0]		
	r/w	r/w	r/w		r/w	r/w	r/w		r/w	r/w	r/w		r/w	r/w	r/w

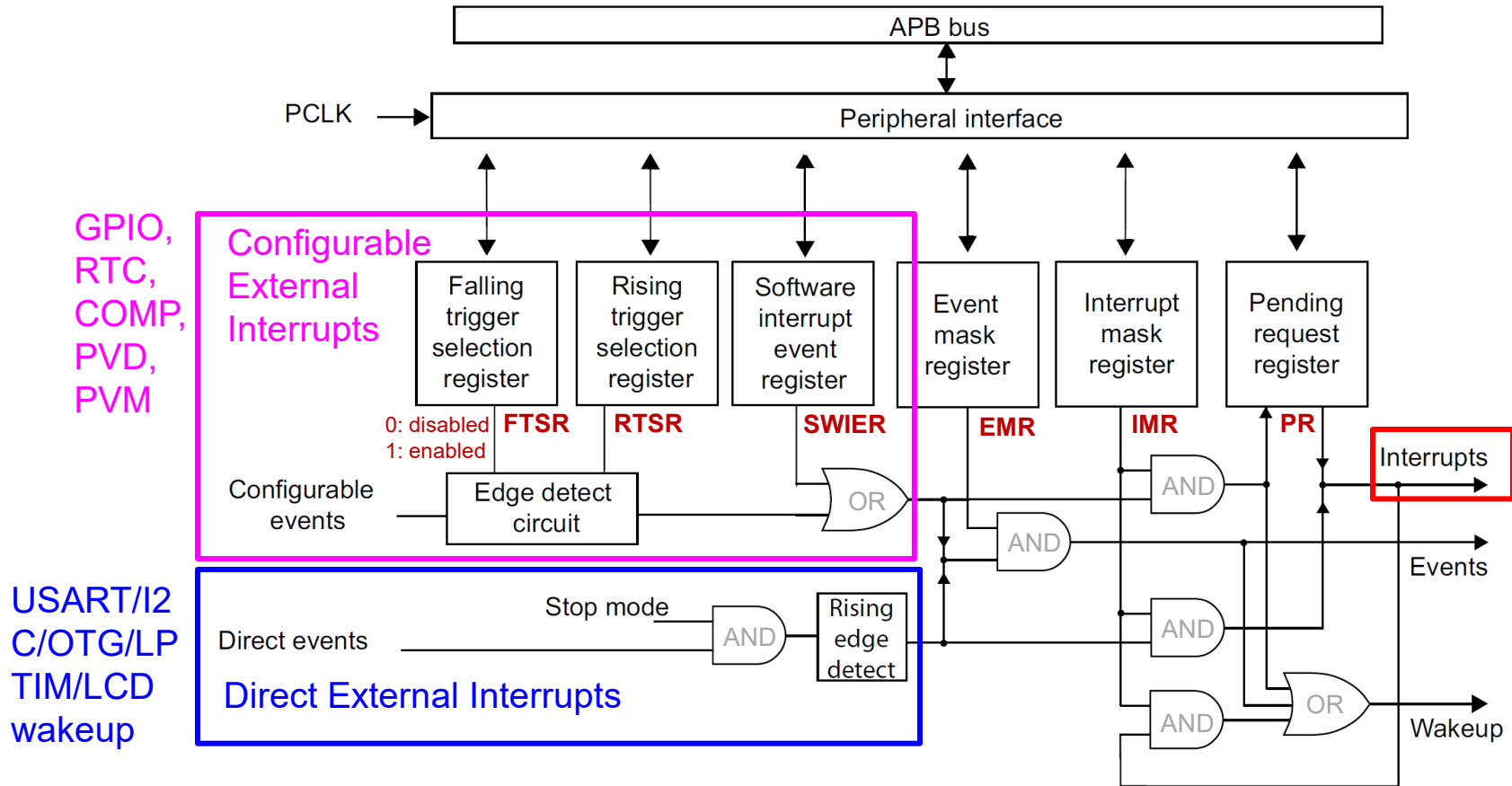
Interrupt Vector Table



Interrupt Number (8 bits)	Memory Address of ISR (32 bits)
...	...
EXTI0_IRQn = 6	EXTI0_IRQHandler
EXTI1_IRQn = 7	EXTI1_IRQHandler
EXTI2_IRQn = 8	EXTI2_IRQHandler
EXTI3_IRQn = 9	EXTI3_IRQHandler
EXTI4_IRQn = 10	EXTI4_IRQHandler
...	...
EXTI9_5_IRQn = 23	EXTI9_5_IRQHandler
...	...
EXTI15_10_IRQn = 40	EXTI15_10_IRQHandler
...	...

first 5 dedicated external interrupts: EXTI 0, EXTI 1, EXTI 2, EXTI 3, and, EXTI 4

External Interrupt (EXTI) Controller



EXTI_RTSR1 (Rising trigger selection register 1)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	RT22	RT21	RT20	RT19	RT18	Res.	RT16
									r/w	r/w	r/w	r/w	r/w		r/w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RT15	RT14	RT13	RT12	RT11	RT10	RT9	RT8	RT7	RT6	RT5	RT4	RT3	RT2	RT1	RT0
r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Bits 31:23 Reserved, must be kept at reset value.

Bits 22:18 **RTx**: Rising trigger event configuration bit of line x (x = 22 to 18)

0: Rising trigger disabled (for Event and Interrupt) for input line

1: Rising trigger enabled (for Event and Interrupt) for input line

Bit 17 Reserved, must be kept at reset value.

Bits 16:0 **RTx**: Rising trigger event configuration bit of line x (x = 16 to 0)

0: Rising trigger disabled (for Event and Interrupt) for input line

1: Rising trigger enabled (for Event and Interrupt) for input line

EXTI_RTSR2 (Rising trigger selection register 2)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	RT38	RT37	RT36	RT35	Res.	Res.	Res.
									rw	rw	rw	rw			

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:3 **RTx**: Rising trigger event configuration bit of line x (x = 35 to 38)

0: Rising trigger disabled (for Event and Interrupt) for input line

1: Rising trigger enabled (for Event and Interrupt) for input line

Bits 2:0 Reserved, must be kept at reset value.

EXTI_FTSR1 (Falling trigger selection register 1)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	FT22	FT21	FT20	FT19	FT18	Res.	FT16
									r/w	r/w	r/w	r/w	r/w		r/w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FT15	FT14	FT13	FT12	FT11	FT10	FT9	FT8	FT7	FT6	FT5	FT4	FT3	FT2	FT1	FT0
r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Bits 31:23 Reserved, must be kept at reset value.

Bits 22:18 **FTx**: Falling trigger event configuration bit of line x (x = 22 to 18)

0: Falling trigger disabled (for Event and Interrupt) for input line

1: Falling trigger enabled (for Event and Interrupt) for input line

Bit 17 Reserved, must be kept at reset value.

Bits 16:0 **FTx**: Falling trigger event configuration bit of line x (x = 16 to 0)

0: Falling trigger disabled (for Event and Interrupt) for input line

1: Falling trigger enabled (for Event and Interrupt) for input line

EXTI_FTSR2 (Falling trigger selection register 2)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	FT38	FT37	FT36	FT35	Res.	Res.	Res.
									rw	rw	rw	rw			

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:3 **FTx**: Falling trigger event configuration bit of line x (x = 35 to 38)

0: Falling trigger disabled (for Event and Interrupt) for input line

1: Falling trigger enabled (for Event and Interrupt) for input line

Bits 2:0 Reserved, must be kept at reset value.

EXTI_IMR1 (Interrupt mask register 1)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
IM31	IM30	IM29	IM28	IM27	IM26	IM25	IM24	IM23	IM22	IM21	IM20	IM19	IM18	IM17	IM16
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IM15	IM14	IM13	IM12	IM11	IM10	IM9	IM8	IM7	IM6	IM5	IM4	IM3	IM2	IM1	IM0
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:0 **IMx**: Interrupt Mask on line x (x = 31 to 0)

0: Interrupt request from Line x is masked

1: Interrupt request from Line x is not masked

Note: The reset value for the direct lines (line 17, lines from 23 to 34, line 39) is set to '1' in order to enable the interrupt by default.

EXTI_IMR2 (Interrupt mask register 2)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	IM39	IM38	IM37	IM36	IM35	IM34	IM33	IM32
								rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:8 Reserved, must be kept at reset value

Bits 7:0 **IMx**: Interrupt mask on line x (x = 39 to 32)

0: Interrupt request from line x is masked

1: Interrupt request from line x is not masked

Note: The reset value for the direct lines (line 17, lines from 23 to 34, line 39) is set to '1' in order to enable the interrupt by default.

EXTI_PR1 (Pending Register 1)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PIF22	PIF21	PIF20	PIF19	PIF18	Res.	PIF16
									rc_w1	rc_w1	rc_w1	rc_w1	rc_w1		rc_w1
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PIF15	PIF14	PIF13	PIF12	PIF11	PIF10	PIF9	PIF8	PIF7	PIF6	PIF5	PIF4	PIF3	PIF2	PIF1	PIF0
rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1

Bits 31:23 Reserved, must be kept at reset value.

Bits 22:18 **PIFx**: Pending interrupt flag on line x (x = 22 to 18)

0: No trigger request occurred

1: Selected trigger request occurred

This bit is set when the selected edge event arrives on the interrupt line. This bit is cleared by writing a '1' to the bit.

Bit 17 Reserved, must be kept at reset value.

Bits 16:0 **PIFx**: Pending interrupt flag on line x (x = 16 to 0)

0: No trigger request occurred

1: Selected trigger request occurred

This bit is set when the selected edge event arrives on the interrupt line. This bit is cleared by writing a '1' to the bit.

EXTI_PR2 (Pending Register 2)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PIF38	PIF37	PIF36	PIF35	Res.	Res.	Res.
									rc_w1	rc_w1	rc_w1	rc_w1			

Bits 31:8 Reserved, must be kept at reset value.

Bit 7 **PIFx**: Pending interrupt flag on line x (x = 35 to 38)

0: No trigger request occurred

1: Selected trigger request occurred

This bit is set when the selected edge event arrives on the interrupt line. This bit is cleared by writing a '1' into the bit.

Bits 2:0 Reserved, must be kept at reset value.

EXTI_SWIER1 (Software interrupt event register 1)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SWI 22	SWI 21	SWI 20	SWI 19	SWI 18	Res.	SWI 16
									rw	rw	rw	rw	rw		rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SWI 15	SWI 14	SWI 13	SWI 12	SWI 11	SWI 10	SWI 9	SWI 8	SWI 7	SWI 6	SWI 5	SWI 4	SWI 3	SWI 2	SWI 1	SWI 0
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:23 Reserved, must be kept at reset value.

Bits 22: 18 **SWIx**: Software interrupt on line x (x = 22 o 18)

If the interrupt is enabled on this line in the EXTI_IMR, writing a '1' to this bit when it is at '0' sets the corresponding pending bit in EXTI_PR resulting in an interrupt request generation.

This bit is cleared by clearing the corresponding bit in the EXTI_PR register (by writing a '1' into the bit).

Bit 17 Reserved, must be kept at reset value.

Bits 16:0 **SWIx**: Software interrupt on line x (x = 16 to 0)

If the interrupt is enabled on this line in the EXTI_IMR, writing a '1' to this bit when it is at '0' sets the corresponding pending bit in EXTI_PR resulting in an interrupt request generation.

This bit is cleared by clearing the corresponding bit of EXTI_PR (by writing a '1' into the bit).

EXTI_SWIER2 (Software interrupt event register 2)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SWI 38	SWI 37	SWI 36	SWI 35	Res.	Res.	Res.
									rw	rw	rw	rw			

Bits 31:8 Reserved, must be kept at reset value.

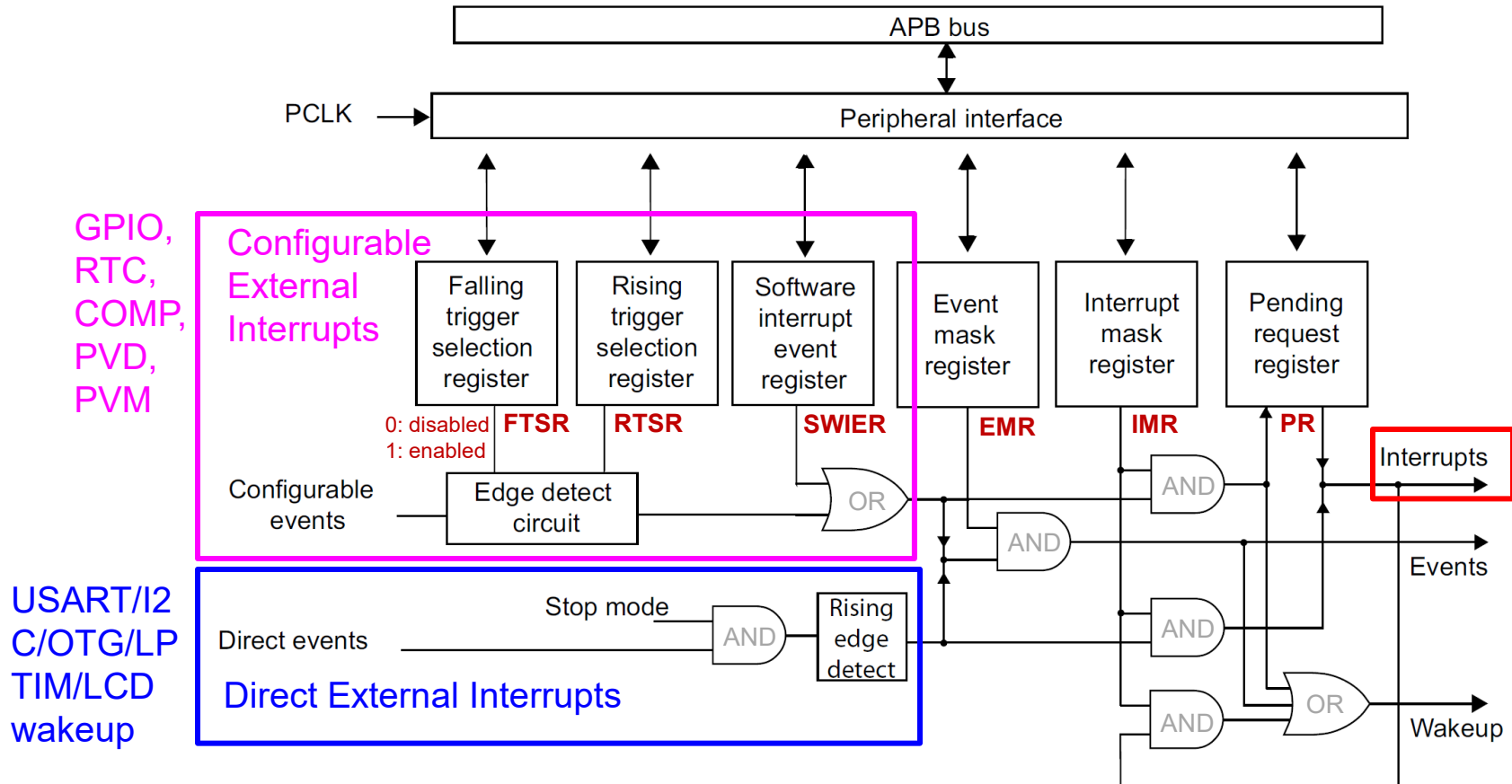
Bit 7 **SWIx**: Software interrupt on line x (x = 35 to 38)

If the interrupt is enabled on this line in EXTI_IMR, writing a '1' to this bit when it is at '0' sets the corresponding pending bit of EXTI_PR resulting in an interrupt request generation.

This bit is cleared by clearing the corresponding bit of EXTI_PR (by writing a '1' to the bit).

Bits 2:0 Reserved, must be kept at reset value.

External Interrupt (EXTI) Controller

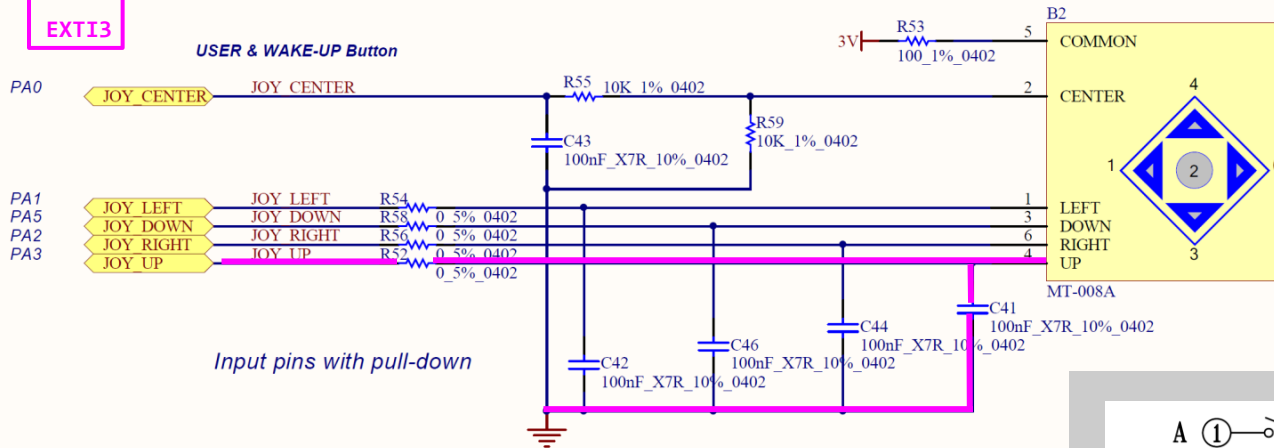


STM32L4 Discovery Kit

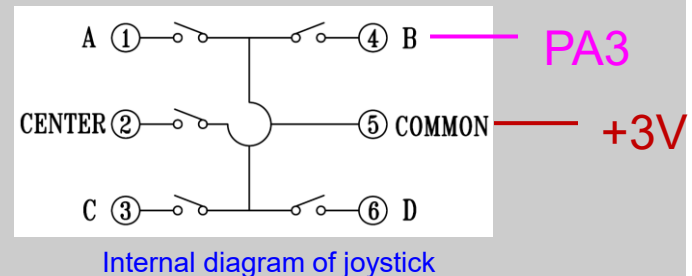
Software goal: If the up button is pressed, turn on LED.

EXTI3

USER & WAKE-UP Button



Input pins with pull-down



Note: PA3 is neither pulled up nor pulled down

```

; Pull PA.3 down internally; Trigger on rising edge
__main      PROC
; Enable GPIO Clock
; RCC->AHB2ENR |= RCC_AHB2ENR_GPIOAEN;

LDR r0, =RCC_BASE
LDR r1, [r0, #RCC_AHB2ENR]
ORR r1, r1, #RCC_AHB2ENR_GPIOAEN
STR r1, [r0, #RCC_AHB2ENR]

ENDP

```

```

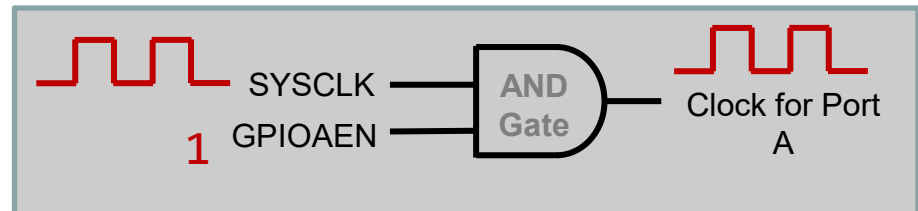
//stm32l476xx.h
#define RCC_AHB2ENR_GPIOAEN
((uint32_t)0x00000001U)

```

AHB2 peripheral clock enable register (AHB2ENR)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	RNG EN	Res.	AESEN
													r/w		r/w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	ADCEN	OTGFS EN	Res.	Res.	Res.	Res.	GPIOH EN	GPIOG EN	GPIOF EN	GPIOE EN	GPIOD EN	GPIOC EN	GPIOB EN	GPIOA EN
		r/w	r/w					r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

1



```
; Pull PA.3 down internally; Trigger on rising edge
__main      PROC
; Enable GPIO Clock
LDR r0, =RCC_BASE
LDR r1, [r0, #RCC_AHB2ENR]
ORR r1, r1, #RCC_AHB2ENR_GPIOAEN
STR r1, [r0, #RCC_AHB2ENR]
; GPIO Mode: Input(00), Output (01),
;           AF(10),   Analog (11, default)
LDR r0, =GPIOA_BASE
LDR r1, [r0, #GPIO_MODER] ; load GPIOA->MODE
BIC r1, r1, #0xC0
STR r1, [r0, #GPIO_MODER]
```

Mode bits = 00

GPIO Pin

Digital Input

Digital Output

Alternative Function

Analog

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MODE15[1:0]		MODE14[1:0]		MODE13[1:0]		MODE12[1:0]		MODE11[1:0]		MODE10[1:0]		MODE9[1:0]		MODE8[1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MODE7[1:0]		MODE6[1:0]		MODE5[1:0]		MODE4[1:0]		MODE3[1:0]		MODE2[1:0]		MODE1[1:0]		MODE0[1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 2y+1:2y **MODEy[1:0]**: Port x configuration bits (y = 0..15)

These bits are written by software to configure the I/O mode.

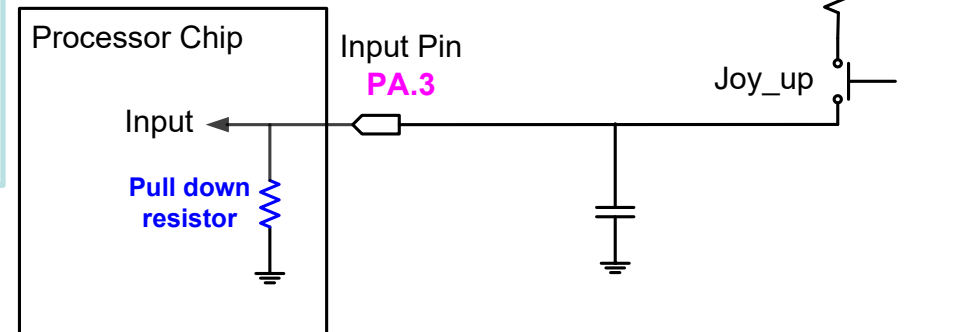
00: Input mode

01: General purpose output mode

10: Alternate function mode

11: Analog mode (reset state)

```
; Pull PA.3 down internally; Trigger on rising edge
__main      PROC
.....
; GPIO Push-Pull: No pull-up, pull-down (00),
; Pull-up (01), Pull-down (10), Reserved (11)
; Pull down
LDR r1, [r0, #GPIO_PUPDR] ; load GPIOA->PUPDR
BIC r1, r1, #0xC0
ORR r1, r1, #0x80
STR r1, [r0, #GPIO_PUPDR] ; store PUPDR
```



31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PUPD15[1:0]		PUPD14[1:0]		PUPD13[1:0]		PUPD12[1:0]		PUPD11[1:0]		PUPD10[1:0]		PUPD9[1:0]		PUPD8[1:0]	
r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PUPD7[1:0]		PUPD6[1:0]		PUPD5[1:0]		PUPD4[1:0]		PUPD3[1:0]		PUPD2[1:0]		PUPD1[1:0]		PUPD0[1:0]	
r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Bits 2y+1:2y **PUPDy[1:0]**: Port x configuration bits (y = 0..15)

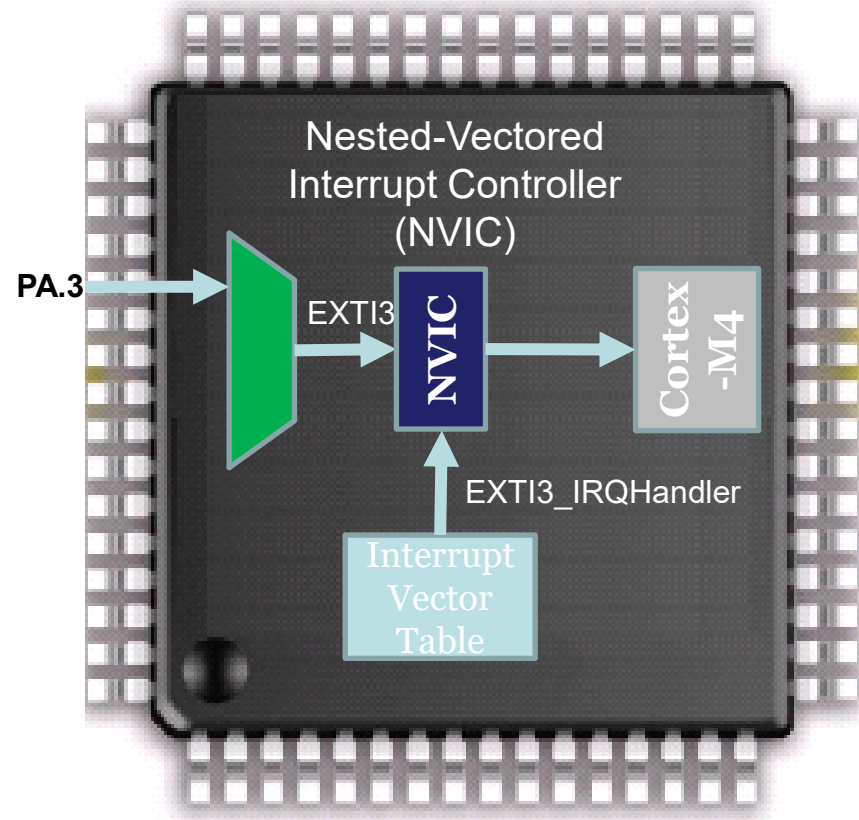
These bits are written by software to configure the I/O pull-up or pull-down

- 00: No pull-up, pull-down
- 01: Pull-up
- 10: Pull-down
- 11: Reserved

```
; Pull PA.3 down internally
; Trigger on rising edge
__main      PROC
... ..

PUSH {LR,R0}
BL config_NVIC_in_C ; Enable Interrupt
POP {LR,R0}
```

```
void config_NVIC_in_C()
{
    NVIC_EnableIRQ(EXTI3_IRQn);
}
```



```
; Pull PA.3 down internally; Trigger on rising edge
__main      PROC
```

```
*****
;Connect External Line to the GPIO
;enable rcc clock
LDR r0, =RCC_BASE
LDR r1, [r0, #RCC_AHB2ENR]
ORR r1, r1, #RCC_APB2ENR_SYSCFGEN
STR r1, [r0, #RCC_AHB2ENR]
;config syscfg_exti
LDR r0, =SYSCFG_BASE
LDR r1, [r0, #SYSCFG_EXTICR0]
BIC r1, #SYSCFG_EXTICR1_EXTI3
ORR r1, #SYSCFG_EXTICR1_EXTI3_PA
STR r1, [r0, #SYSCFG_EXTICR0]
```

Bit 0 **SYSCFGEN**: SYSCFG + COMP + VREFBUF clock enable

Set and cleared by software.

0: SYSCFG + COMP + VREFBUF clock disabled

1: SYSCFG + COMP + VREFBUF clock enabled

Bits 15:12 **EXTI3[3:0]**: EXTI 3 configuration bits

These bits are written by software to select the source input for the EXTI3 external interrupt.

0000: PA[3] pin

0001: PB[3] pin

0010: PC[3] pin

0011: PD[3] pin

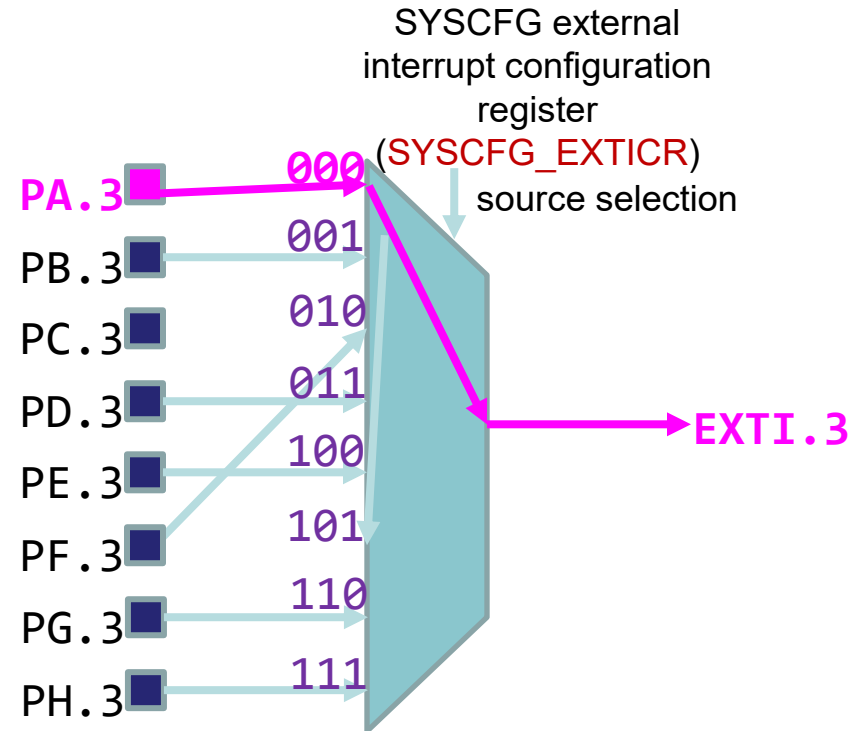
0100: PE[3] pin

0101: PF[3] pin

0110: PG[3] pin

0111: PH[3] pin (only on STM32L496xx/4A6xx devices)

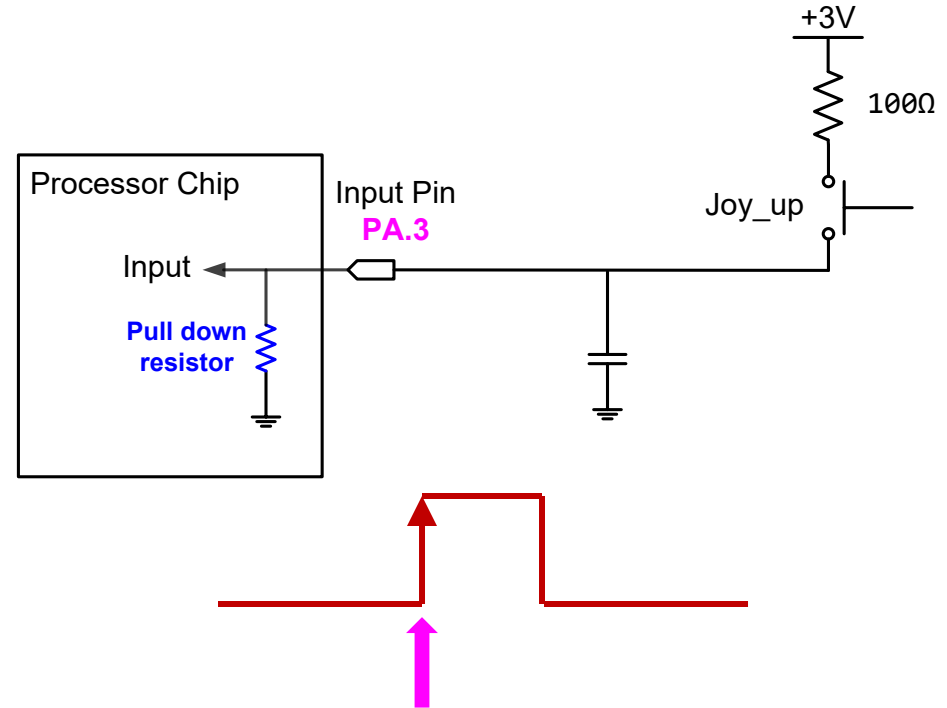
1000: PI[3] pin (only for STM32L496xx/4A6xx devices)



```

__main      PROC
.....
; Rising trigger selection
; 0 = trigger disabled, 1 = trigger enabled
; enable rising edge trigger
LDR r0, =EXTI_BASE
LDR r1, [r0, #EXTI_RTSR1]
ORR r1, #EXTI_RTSR1_RT3
STR r1, [r0, #EXTI_RTSR1]

ENDP
    
```



Rising edge triggers EXTI interrupt.

```
__main      PROC
*****
; Rising trigger selection
; 0 = trigger disabled, 1 = trigger enabled
LDR r0, =EXTI_BASE
LDR r1, [r0, #EXTI_RTSR1]
ORR r1, #EXTI_RTSR1_RT3
STR r1, [r0, #EXTI_RTSR1]

ENDP
```

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res	Res	Res	Res	Res	Res	Res	Res	Res	RT22	RT21	RT20	RT19	RT18	Res	RT16
									r/w	r/w	r/w	r/w	r/w		r/w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RT15	RT14	RT13	RT12	RT11	RT10	RT9	RT8	RT7	RT6	RT5	RT4	RT3	RT2	RT1	RT0
r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Bits 31:23 Reserved, must be kept at reset value.

Bits 22:18 **RTx**: Rising trigger event configuration bit of line x (x = 22 to 18)

0: Rising trigger disabled (for Event and Interrupt) for input line

1: Rising trigger enabled (for Event and Interrupt) for input line

Bit 17 Reserved, must be kept at reset value.

Bits 16:0 **RTx**: Rising trigger event configuration bit of line x (x = 16 to 0)

0: Rising trigger disabled (for Event and Interrupt) for input line

1: Rising trigger enabled (for Event and Interrupt) for input line

Note:

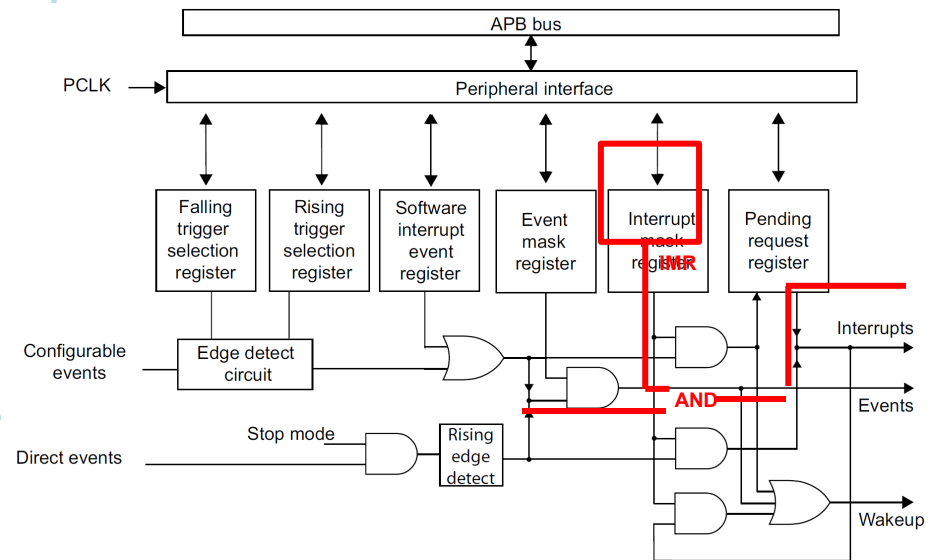
The configurable wakeup lines are edge-triggered. No glitch must be generated on these lines. If a rising edge on a configurable interrupt line occurs during a write operation in the EXTI_RTSR register, the pending bit is not set.

Rising and falling edge triggers can be set for the same interrupt line. In this case, both generate a trigger condition.

```

; Pull PA.3 down internally; Trigger on rising edge
__main      PROC
; Interrupt Mask Register
; 0 = masked, 1 = not masked (enabled)
LDR r0, =EXTI_BASE
LDR r1, [r0, #EXTI_IMR1]
ORR r1, # EXTI_IMR1_IM3
STR r1, [r0, # EXTI_IMR1]

ENDP
    
```



```
; Pull PA.3 down internally; Trigger on rising edge
__main      PROC
.....
; Interrupt Mask Register
; 0 = masked, 1 = not masked (enabled)
LDR r0, =EXTI_BASE
LDR r1, [r0, #EXTI_IMR1]
ORR r1, # EXTI_IMR1_IM3
STR r1, [r0, # EXTI_IMR1]

ENDP
```

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
IM31	IM30	IM29	IM28	IM27	IM26	IM25	IM24	IM23	IM22	IM21	IM20	IM19	IM18	IM17	IM16
r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IM15	IM14	IM13	IM12	IM11	IM10	IM9	IM8	IM7	IM6	IM5	IM4	IM3	IM2	IM1	IM0
r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Bits 31:0 **IMx**: Interrupt Mask on line x (x = 31 to 0)

0: Interrupt request from Line x is masked

1: Interrupt request from Line x is not masked

Note: The reset value for the direct lines (line 17, lines from 23 to 34, line 39) is set to '1' in order to enable the interrupt by default.

```
; Pull PA.3 down internally; Trigger on  
rising edge
```

```
__main      PROC
```

```
.....
```

```
; Interrupt Mask Register
```

```
; 0 = marked, 1 = not masked (enabled)
```

```
LDR r0, =EXTI_BASE
```

```
LDR r1, [r0, #EXTI_IMR1]
```

```
ORR r1, # EXTI_IMR1_IM3
```

```
STR r1, [r0, # EXTI_IMR1]
```

```
stop
```

```
B stop
```

```
ENDP
```

Processor runs a dead loop
forever!



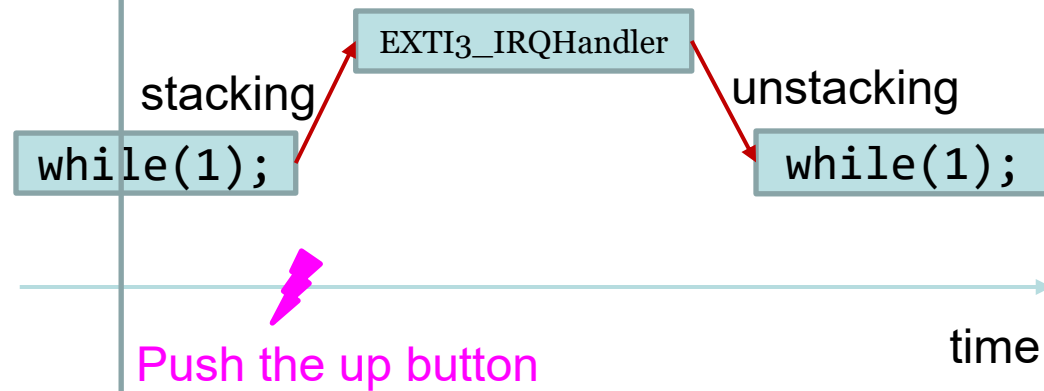
Interrupt Service Routine (ISR)

EXTI3_IRQHandler PROC

```
LDR r0, =EXTI_BASE
LDR r1, [r0, #EXTI_PR1]
AND r1, #EXTI_PR1_PIF3
CMP r1, #EXTI_PR1_PIF3
BNE end
; Toggle LED
; Cleared flag by writing 1
LDR r0, =EXTI_BASE
LDR r1, [r0, #EXTI_PR1]
ORR r1, #EXTI_PR1_PIF3
STR r1, [r0, #EXTI_PR1]
```

end

ENDP



Normal Motors vs. Stepper Motors



Normal Motors

- 2-wire interface
- Current causes shaft to spin
- Imprecise

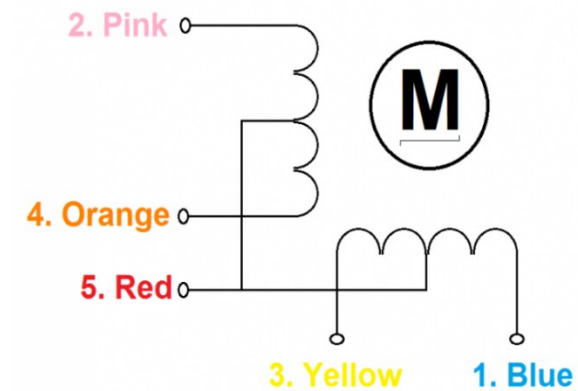


Stepper Motors

- 5-wire interface (usually)
 - Spinning of shaft is driven by magnet phases
 - Precise

Electrical and Computer Engineering

Stepper Motor



Stepper Motor

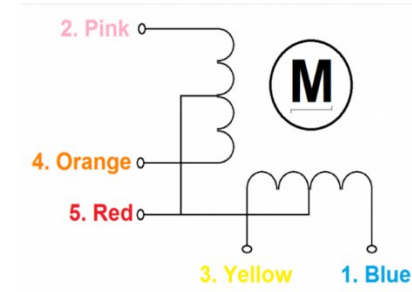
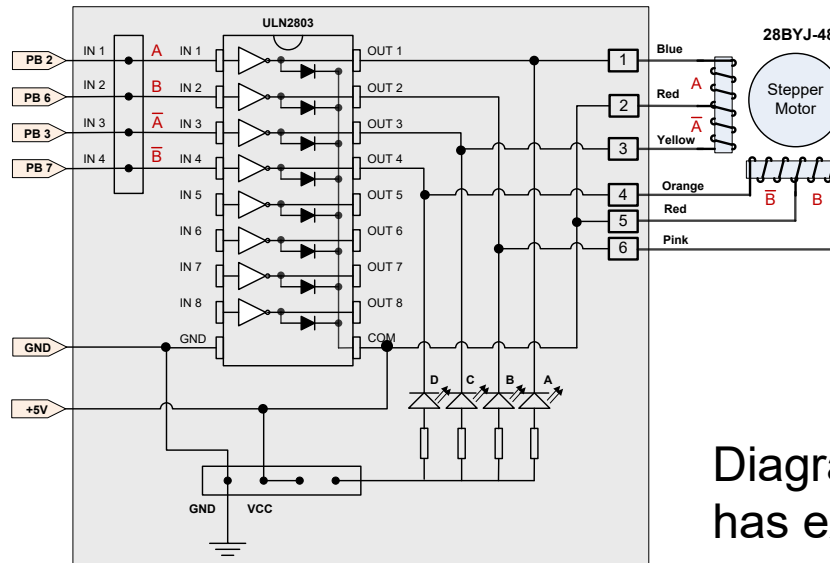
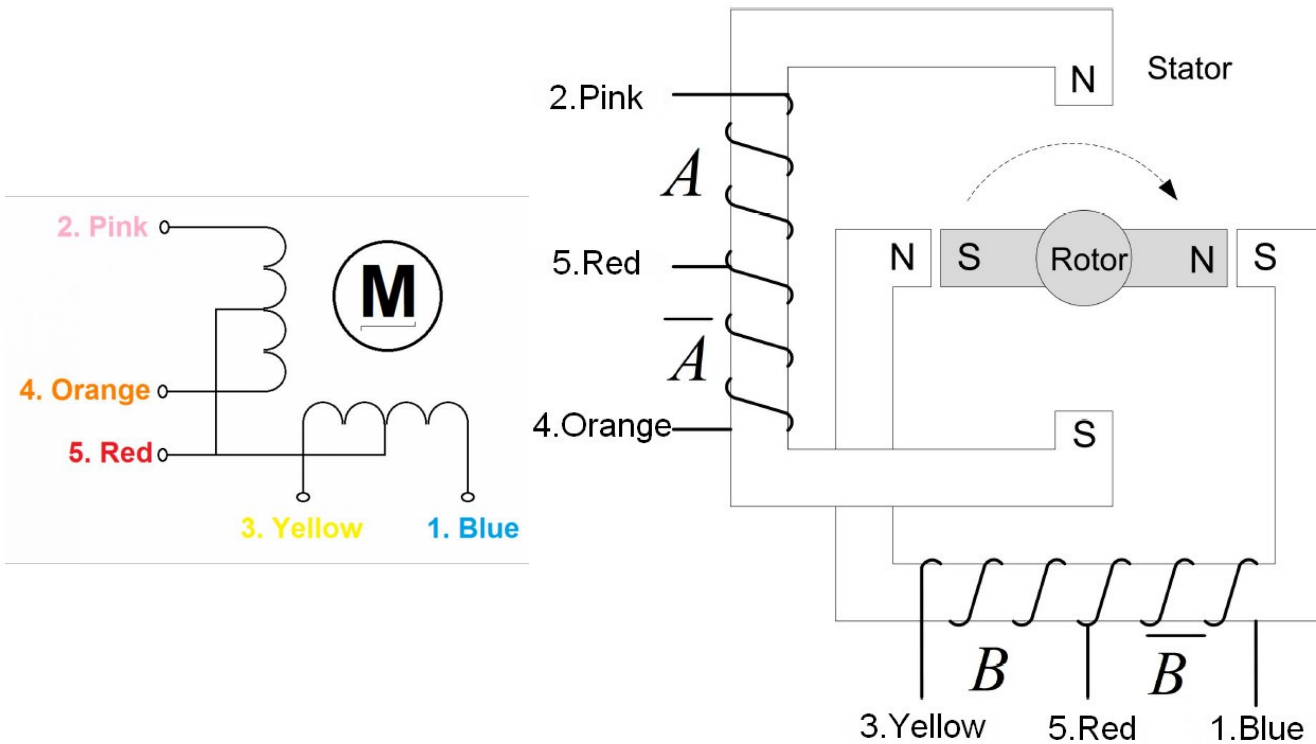


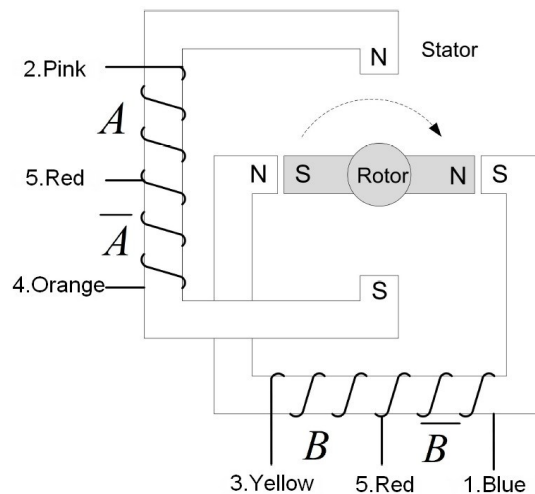
Diagram from lab –
has extra red wire
connection

Electrical and Computer Engineering

Inner Structure



On/Off to magnet activation



If $A = 1$ and $\bar{A} = 0$, the A magnet becomes South and the $\bar{A}$ magnet becomes North.

If $A = 0$ and $\bar{A} = 1$, the A magnet becomes North and the $\bar{A}$ magnet becomes South.

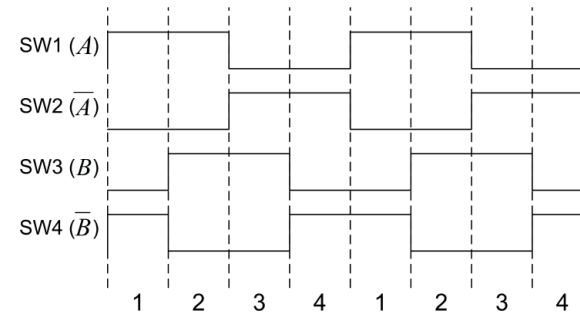
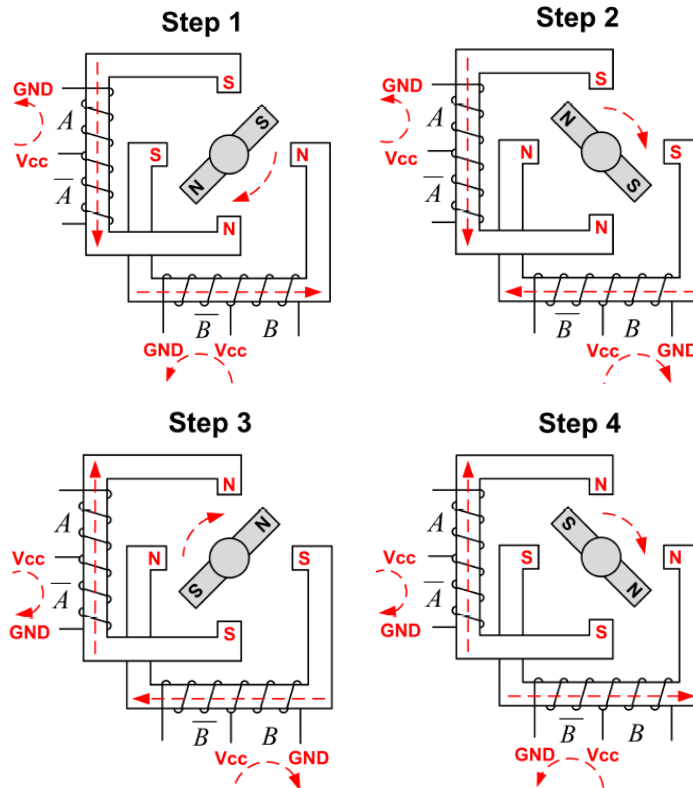
If A and $\bar{A}$ have the same state, the magnet will not be activated.

The same corresponds to B and $\bar{B}$.

Electrical and Computer Engineering

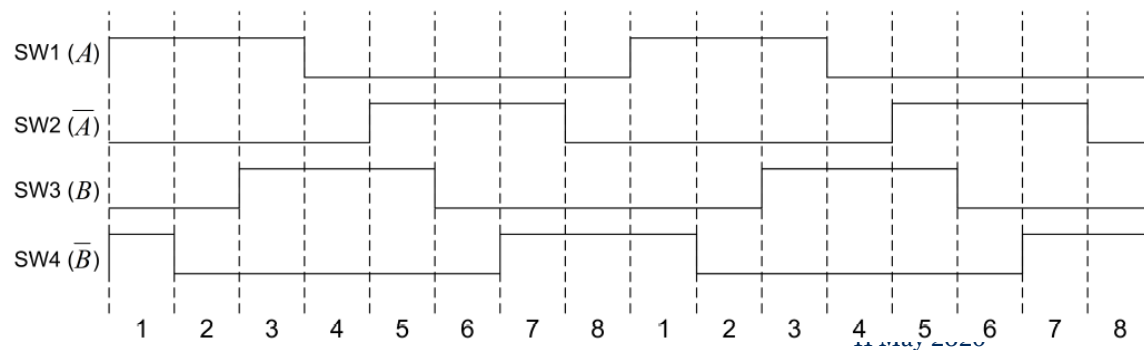
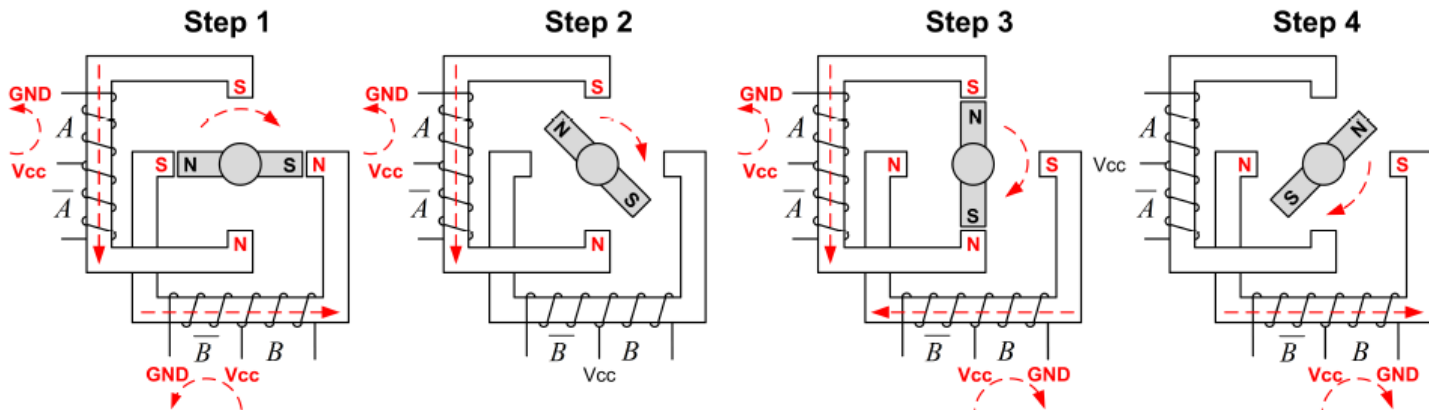
Full-Stepping

Coil activation sequence for clockwise rotation: $A\bar{B}$, AB , $\bar{A}B$, $\bar{A}\bar{B}$



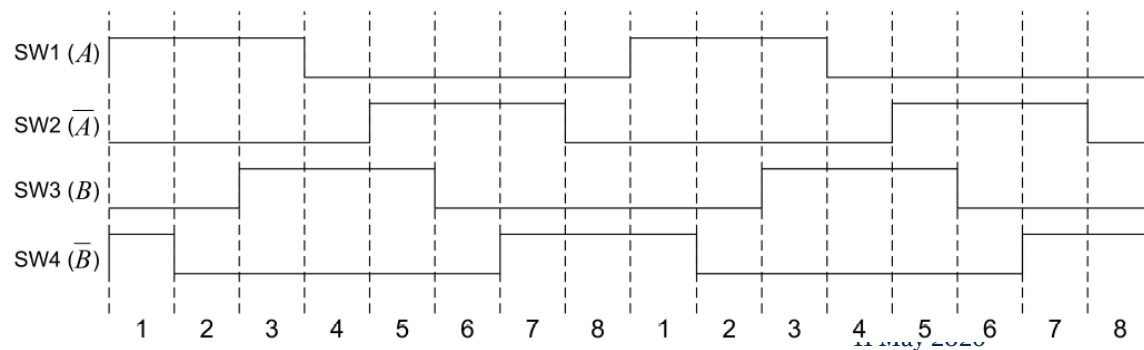
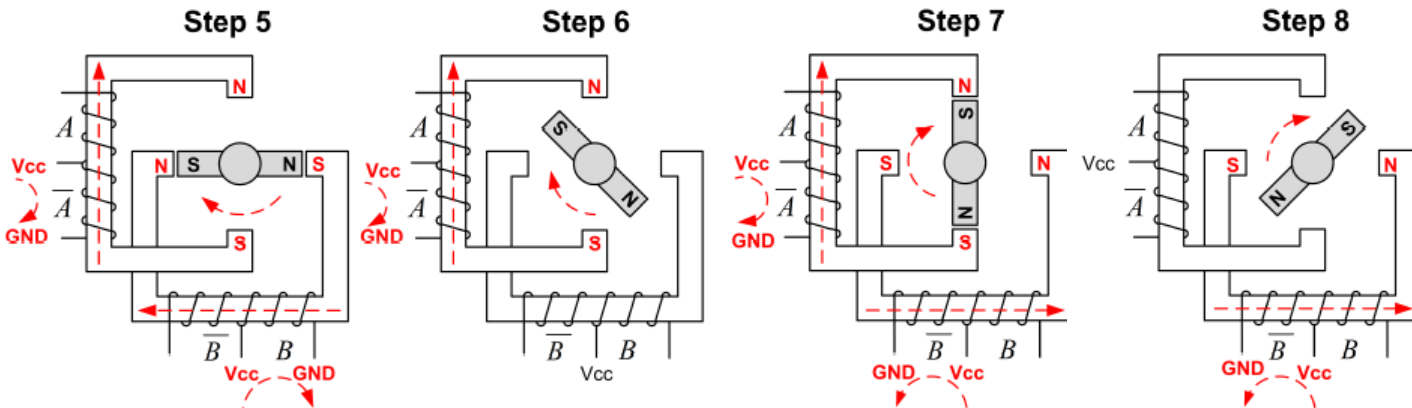
Half-Stepping

Coil activation sequence: $A\bar{B}$, A , AB , B , $\bar{A}B$, $\bar{A}$, $\bar{A}\bar{B}$, $\bar{B}$



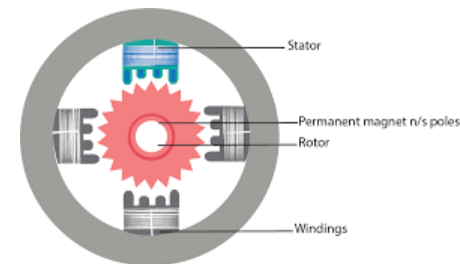
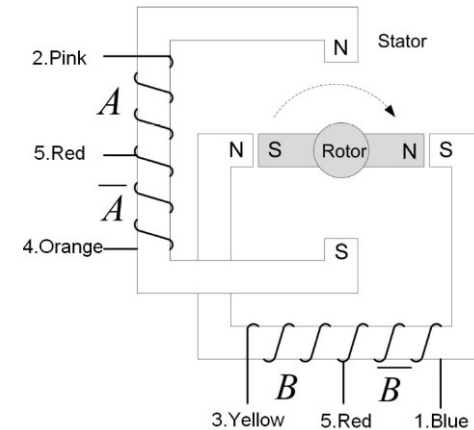
Half-Stepping

Coil activation sequence: $A\bar{B}$, A , AB , B , $\bar{A}B$, $\bar{A}$, $\bar{A}\bar{B}$, $\bar{B}$



Number of Steps for a Revolution

- Rotor
 - 8 pairs of poles, not 1 pair.
- Full-stepping
 - Internal motor: $4 * 8 = 32$ steps per revolution
- Half-stepping
 - Internal motor: $8 * 8 = 64$ steps per revolution



<https://openlabpro.com/guide/stepper-motor-basics/>

Number of Steps for a Revolution

- Rotor
 - 8 pairs of poles, not 1 pair.
- Full-stepping
 - Internal motor: $4 * 8 = 32$ steps per revolution
 - Gear ratio: 64
 - Total: $32 * 64 = 2048$ steps per revolution
- Half-stepping
 - Internal motor: $8 * 8 = 64$ steps per revolution
 - Gear ratio: 64
 - Total: $64 * 64 = 4096$ steps per revolution

